University of Greater Manchester

| **Mini Project 3: Northwest Architecture Studio Network** | | **Team (20%)** |
|---|---|---|
| **Networking Fundamentals:** LAN and external/cloud network basics, IP addressing, protocols, and client-server architecture built and tested in Filius | | |
| **Learning Outcome** | **How this mini project contributes** | **Typical Evidence** |
| **LO1:** Demonstrate foundational knowledge of core computing systems and practices | Demonstrate understanding of LAN and external-network basics, IP addressing, common protocols (DNS, HTTP, ICMP/ping, TCP/IP) and client-server architecture by building and testing a working two-network simulation in Filius. | Topology screenshot; completed IP addressing table; router-interface and DNS-record screenshots; protocol table. |
| **LO2:** Apply evidence-based learning and study strategies to personal academic work | Apply a structured process - plan the addressing, build the topology, configure services, test methodically, then diagnose and fix any faults. | Practice-task evidence; "who did what" task-allocation grid; testing table; troubleshooting log of a fault found and fixed. |
| **LO3:** Produce clearly structured and correctly referenced academic outputs to explain computing concepts in written, oral, and collaborative formats | Produce a clear GitHub/Markdown portfolio section explaining the network, the protocols and the client-server model, with screenshots, tables and team-contribution evidence. | GitHub portfolio section; protocol and client-server explanations; ping/browser test evidence; team contribution record. |

## Background

Northwest Architecture Studio Ltd is refreshing its IT infrastructure. The firm has staff PCs for administrative, project-management, CAD/design and 3D-visualisation roles. As a team, you must model a working office LAN and a connection to an external cloud/ISP network in Filius software, assign IP addresses, set up internal and external services, test the network, and explain the networking principles.

You do not need advanced subnetting, NAT, firewall rules or professional router configuration. The focus is on demonstrating correct foundational understanding through a working simulation and a clear explanation.

## Indicative content mapping

| **Indicative content area** | **Where it is covered** | **Typical evidence** |
|---|---|---|
| Internet / external network basics | The office LAN connects through a router to an external ISP/cloud network. | Topology screenshot; explanation of router, gateway and external network. |
| LAN basics | Staff PCs, office server and DNS server connect through an internal office switch. | Office LAN diagram; explanation of switch and LAN. |
| IP addressing | Assign static IP addresses to office and external devices using two /24 networks. | Completed IP addressing table; screenshots of device settings. |
| Overview of protocols | Explain DNS, HTTP, ICMP/ping, TCP/IP and gateway routing at overview level. | Protocol table and testing evidence. |
| Client-server architecture | Client PCs request internal and external web pages from servers. | Browser screenshots and explanation of request/response. |

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# Network design

## Required network model

| Network area | Network address | Purpose |
|---|---|---|
| Office LAN | 192.168.10.0/24 | Private internal office network for staff PCs and office services. |
| External / ISP / Cloud | 203.0.113.0/24 | A simulated outside network representing ISP/cloud access. |

## Important modelling note

In a real organisation the router would connect to a modem or fibre ONT and then the ISP. Here the external switch and ISP/cloud server represent the outside network. This keeps the simulation simple while still showing the key principle: a router connects different networks.

## Devices to create in Filius

| Filius device | Rename as | Purpose |
|---|---|---|
| Router (2 connections) | Office Router | Connects the office LAN to the external/cloud network. |
| Switch | Office Switch | Connects internal office devices. |
| Switch | External / ISP Switch | Represents the external network connection. |
| Computer / Notebook | Admin PC | Office user/client device (Administrative role). |
| Computer / Notebook | Project PC | Office user/client device (Project Manager role). |
| Computer / Notebook | CAD PC | Office user/client device (CAD/Design role). |
| Computer / Notebook | Visual PC | Office user/client device (3D Visualisation role). |
| Computer | DNS Server | Provides name resolution for internal and external names. |
| Computer | Office Server | Hosts the internal company intranet page. |
| Computer | ISP / Cloud Server | Hosts an external cloud service page. |

## IP addressing plan

| Device | IP address | Subnet mask | Gateway | DNS |
|---|---|---|---|---|
| Router int.1 (Office side) | 192.168.10.1 | 255.255.255.0 | – | – |
| Router int.2 (External side) | 203.0.113.1 | 255.255.255.0 | – | – |
| DNS Server | 192.168.10.2 | 255.255.255.0 | 192.168.10.1 | 192.168.10.2 |
| Office Server | 192.168.10.10 | 255.255.255.0 | 192.168.10.1 | 192.168.10.2 |
| Admin PC | 192.168.10.101 | 255.255.255.0 | 192.168.10.1 | 192.168.10.2 |
| Project PC | 192.168.10.111 | 255.255.255.0 | 192.168.10.1 | 192.168.10.2 |
| CAD PC | 192.168.10.121 | 255.255.255.0 | 192.168.10.1 | 192.168.10.2 |
| Visual PC | 192.168.10.131 | 255.255.255.0 | 192.168.10.1 | 192.168.10.2 |
| ISP / Cloud Server | 203.0.113.10 | 255.255.255.0 | 203.0.113.1 | 192.168.10.2 |

**Why two networks?**

Office devices use 192.168.10.x addresses; the external/cloud server uses a different 203.0.113.x address. The router sits between the two networks and allows communication between them.

**DNS records to create**

| DNS name | Points to | Meaning |
|---|---|---|
| intranet.archstudio.local | 192.168.10.10 | Internal office intranet server. |
| cloud.archstudio.example | 203.0.113.10 | External/cloud service server. |

**Web pages.** On the Office Server and on the ISP / Cloud Server, use Text Editor to save a simple index.html inside the webserver folder (a heading and one line of text is enough. The aim is to prove a client can request a page from a server, not web design), then start each Webserver.

## Step-by-step build instructions

1. Open Filius in design mode. Add one router with 2 connections; rename it Office Router.
2. Add one Office Switch and one External / ISP Switch.
3. Connect router connection 1 to the Office Switch, and connection 2 to the External / ISP Switch.
4. Add and rename the office devices: Admin PC, Project PC, CAD PC, Visual PC, DNS Server and Office Server; connect all of them to the Office Switch.
5. Add one external computer, rename it ISP / Cloud Server, and connect it to the External / ISP Switch.
6. Configure router interface 1 as 192.168.10.1 / 255.255.255.0 and interface 2 as 203.0.113.1 / 255.255.255.0.

   ***Router note:*** *Because a single router connects the two networks directly, Filius already knows how to route between them — no forwarding table or "Automatic Routing" setting needs configuring. Just set the two interface IPs above and give every device its gateway. (If you are on an older Filius build and routing fails, open the router and make sure IP forwarding / "Automatic Routing" is enabled.)*
7. Configure every device using the IP addressing table.
8. In simulation mode, install Command Line on the client PCs for ping testing.
9. Install DNS Server software on the DNS Server and add both DNS records.
10. Install Webserver and Text Editor on the Office Server and ISP / Cloud Server; create and save the two index.html pages; start both web servers.
11. Install Web Browser on the CAD PC and complete the tests in the testing table.
12. Take screenshots as evidence and write your explanations.

## Required tests (from the CAD PC)

| Test | Command / action | What it proves |
|---|---|---|
| LAN PC-to-PC | ping 192.168.10.101 | Office LAN devices communicate through the switch. |
| PC to Office Server | ping 192.168.10.10 | Client can reach the internal server. |

| | | |
|---|---|---|
| PC to DNS Server | ping 192.168.10.2 | Client can reach the DNS service. |
| Internal web by IP | http://192.168.10.10 | HTTP client-server works internally. |
| Internal web by name | http://intranet.archstudio.local | DNS resolves the internal name to its IP. |
| External server ping | ping 203.0.113.10 | Router connects the office LAN to the external network. |
| External web by IP | http://203.0.113.10 | Client can access an external service. |
| External web by name | http://cloud.archstudio.example | DNS resolves the external service name. |

## Protocols and concepts to explain (at least five)

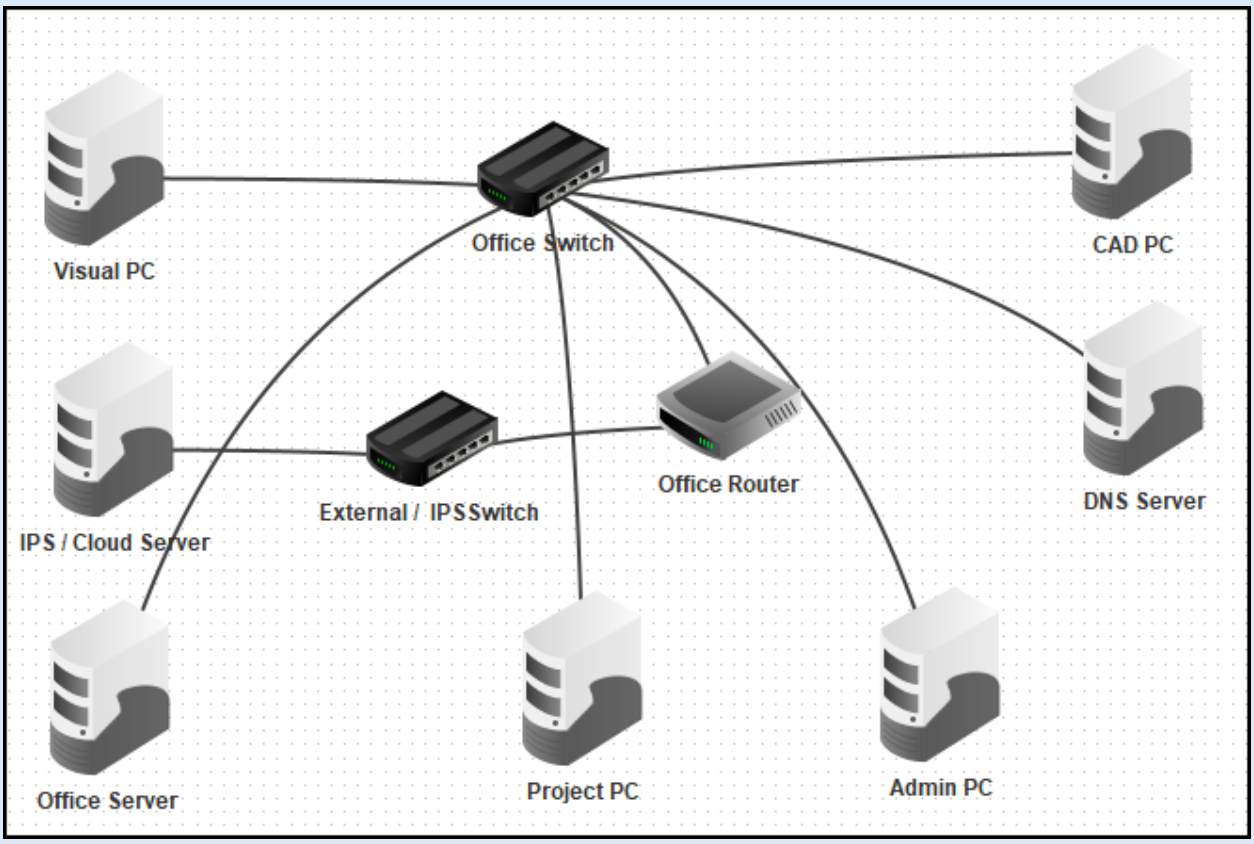
| Concept | Where it appears |
|---|---|
| TCP/IP | All device communication |
| IP addressing | Every device has a unique IP |
| Subnet mask | 255.255.255.0 on both networks |
| Gateway | Client PCs use 192.168.10.1 |
| DNS | intranet.archstudio.local etc. |
| HTTP | Opening pages in the browser |
| ICMP / ping | Ping tests |
| Client-server | CAD PC → Office/Cloud Server |
| Switch vs router | Office/External switch; Office Router |

## Evidence to submit, team roles & contribution

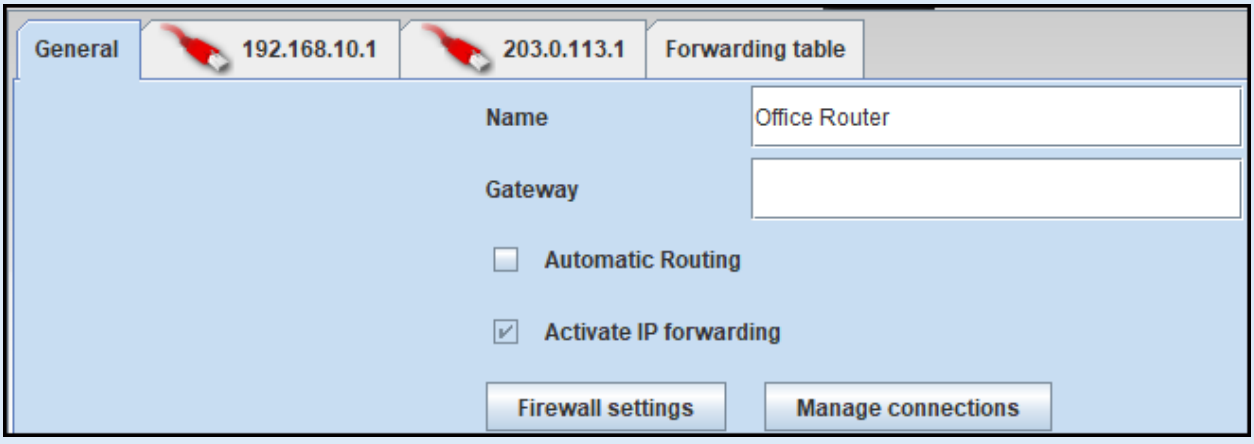
**Required evidence.** Topology screenshot (office LAN, router, external switch, ISP/cloud server); completed IP addressing table; router-interface screenshot; DNS-records screenshot; internal and external web pages opening from the CAD PC; at least three successful ping tests; a protocol table explaining 5+ concepts; a short explanation of the LAN vs external network and of the client-server model.

**Working as a team.** Each member owns a clearly defined part of the build and its write-up, and everyone contributes to testing and the final explanation. Record who did what in a contribution grid so an individual percentage can be awarded.

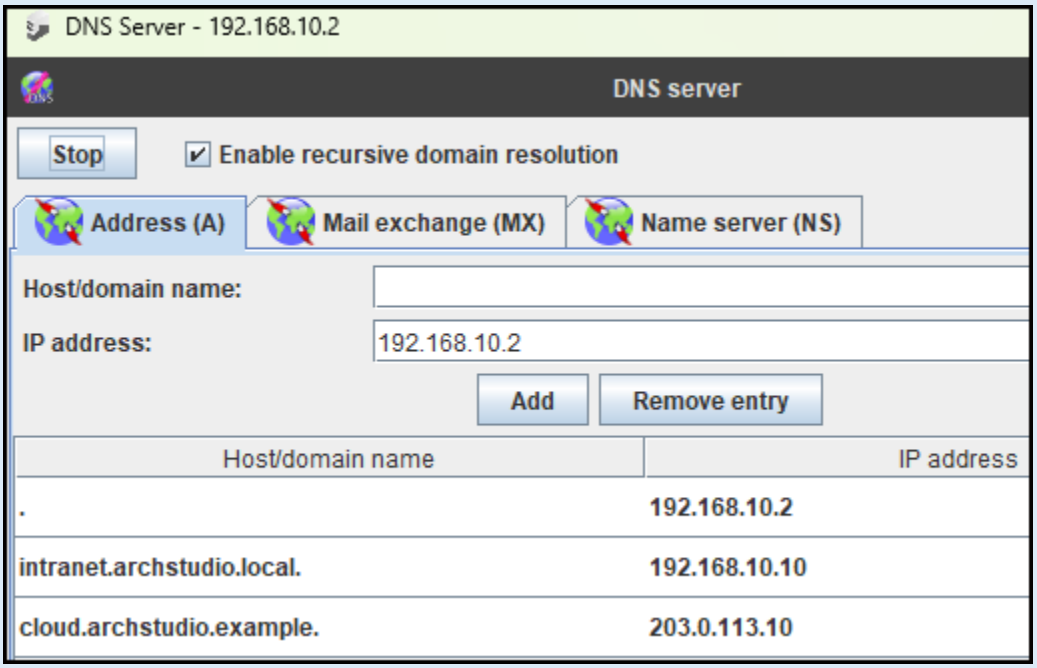
The following network devices have been created in Filius software and assigned each device with an IP address according to the table.




**Office Router:**




**DNS Record**




**Internal webpage opening from( CAD PC)**

**Internal webpage by IP Address**




**Internal webpage by name**




**Data exchange**

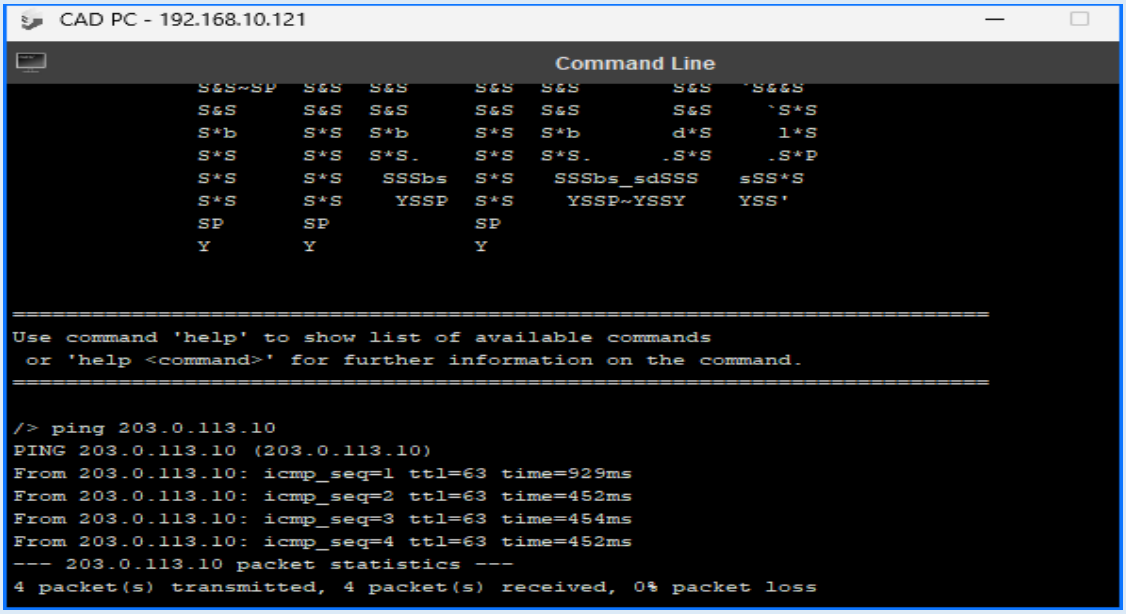
Data exchange

Office Router - 192.168.10.1 X | CAD PC - 192.168.10.121 X

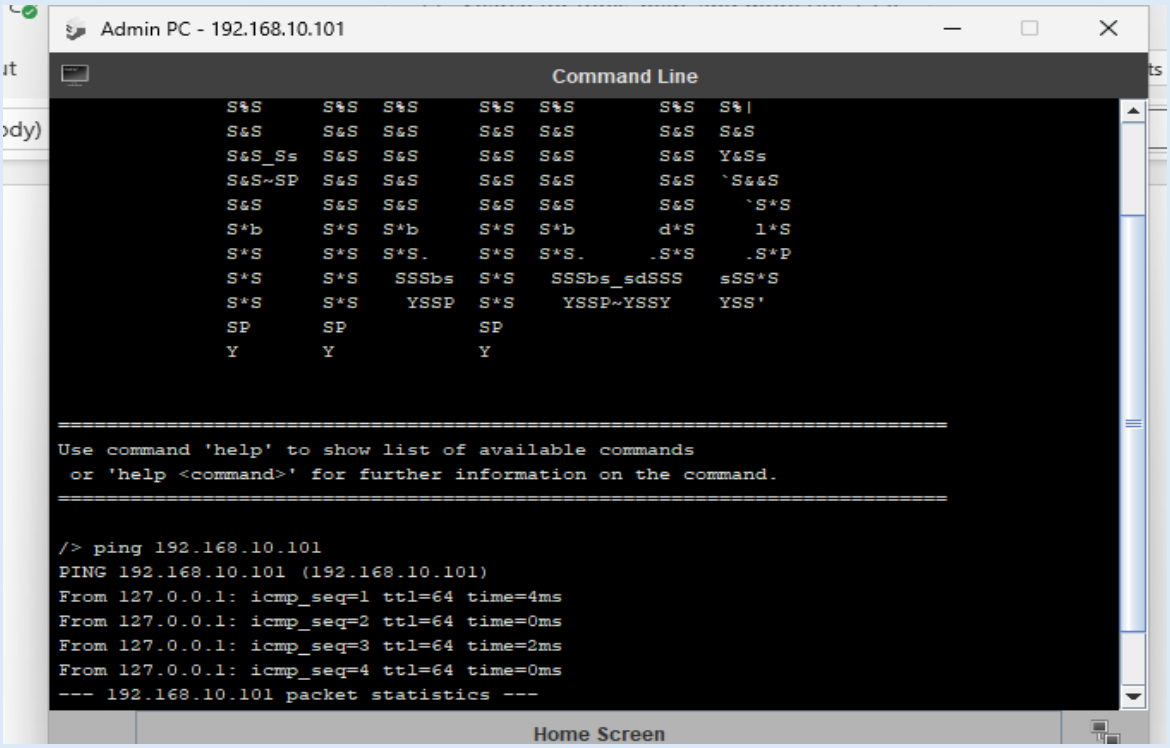
| No. | Time | Source | Destination | Protocol | Layer | Comment / Details |
|---|---|---|---|---|---|---|
| 14 | 10:59:49.446 | 192.168... | 192.168... | ARP | Internet | Search for MAC 192.168.10.2 [op=REQUEST, sender=E8:5D:E3:BD:C2:74\|192.168.10.121, target=FF:FF:FF:FF:FF:FF\|192.168.10.2] |
| 15 | 10:59:49.673 | 192.168... | 192.168... | ARP | Internet | MAC is F4:1B:CC:AC:75:4D [op=REPLY, sender=F4:1B:CC:AC:75:4D\|192.168.10.2, target=E8:5D:E3:BD:C2:74\|192.168.10.121] |
| 16 | 10:59:49.673 | 192.168... | 192.168... | DNS | Application | ID=37082 QR=0 RCODE=0 QDCOUNT=1 ANCOUNT=0 NSCOUNT=0 ARCOUNT=0 intranet.archstudio.loca. A IN |
| 17 | 10:59:49.906 | 192.168... | 192.168... | DNS | Application | ID=37082 QR=1 RCODE=0 QDCOUNT=0 ANCOUNT=0 NSCOUNT=0 ARCOUNT=0 |
| 18 | 10:59:54.698 | 192.168... | 192.168... | DNS | Application | ID=42329 QR=0 RCODE=0 QDCOUNT=1 ANCOUNT=0 NSCOUNT=0 ARCOUNT=0 intranet.archstudio.loca. A IN |
| 19 | 10:59:54.930 | 192.168... | 192.168... | DNS | Application | ID=42329 QR=1 RCODE=0 QDCOUNT=0 ANCOUNT=0 NSCOUNT=0 ARCOUNT=0 |
| 20 | 11:00:01.665 | 192.168... | 192.168... | DNS | Application | ID=50128 QR=0 RCODE=0 QDCOUNT=1 ANCOUNT=0 NSCOUNT=0 ARCOUNT=0 intranet.archstudio.local. A IN |
| 21 | 11:00:01.897 | 192.168... | 192.168... | DNS | Application | ID=50128 QR=1 RCODE=0 QDCOUNT=0 ANCOUNT=1 NSCOUNT=0 ARCOUNT=0 intranet.archstudio.local. A 3600 192.168.10.10 |
| 22 | 11:00:01.898 | 192.168... | 192.168... | TCP | Transport | SYN, SEQ: 11,000,000 |
| 23 | 11:00:02.124 | 192.168... | 192.168... | TCP | Transport | SYN, SEQ: 10,000,000, ACK: 11,000,001 |
| 24 | 11:00:02.125 | 192.168... | 192.168... | TCP | Transport | SEQ: 11,000,001, ACK: 10,000,001 |
| 25 | 11:00:02.199 | 192.168... | 192.168... | HTTP | Application | GET / HTTP/1.1 Host: intranet.archstudio.local |
| 26 | 11:00:02.425 | 192.168... | 192.168... | TCP | Transport | SEQ: 10,000,001, ACK: 11,000,048 |
| 27 | 11:00:02.483 | 192.168... | 192.168... | HTTP | Application | HTTP/1.1 200 OK Content-type: text/html <!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport" con... |
| 28 | 11:00:02.483 | 192.168... | 192.168... | TCP | Transport | SEQ: 11,000,048, ACK: 10,000,337 |
| 29 | 11:00:02.539 | 192.168... | 192.168... | TCP | Transport | FIN, SEQ: 11,000,048 |
| 30 | 11:00:02.765 | 192.168... | 192.168... | TCP | Transport | SEQ: 10,000,337, ACK: 11,000,049 |
| 31 | 11:00:02.823 | 192.168... | 192.168... | TCP | Transport | FIN, SEQ: 10,000,337 |
| 32 | 11:00:02.823 | 192.168... | 192.168... | TCP | Transport | SEQ: 11,000,049, ACK: 10,000,338 |

## Ping test for client PC

### External server ping



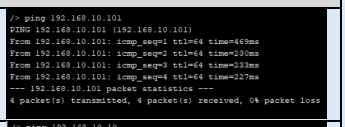
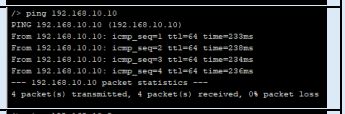
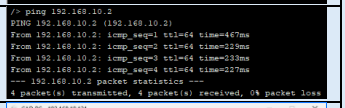
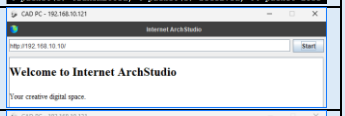
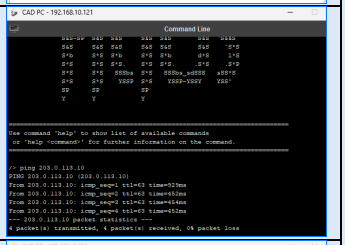
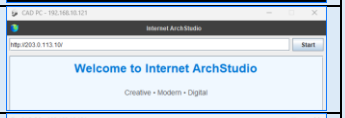
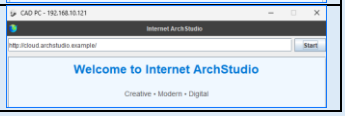

### Admin PC ping




**DNS server ping**

```
/> ping 192.168.10.2
PING 192.168.10.2 (192.168.10.2)
From 192.168.10.2: icmp_seq=1 ttl=64 time=467ms
From 192.168.10.2: icmp_seq=2 ttl=64 time=229ms
From 192.168.10.2: icmp_seq=3 ttl=64 time=233ms
From 192.168.10.2: icmp_seq=4 ttl=64 time=227ms
--- 192.168.10.2 packet statistics ---
4 packet(s) transmitted, 4 packet(s) received, 0% packet loss
```

TEST (from the CAD PC)

| Test | Command / action | What it proves | Result screenshot |
|---|---|---|---|
| LAN PC-to-PC | ping 192.168.10.101 | Office LAN devices communicate through the switch. |  |
| PC to Office Server | ping 192.168.10.10 | Client can reach the internal server. |  |
| PC to DNS Server | ping 192.168.10.2 | Client can reach the DNS service. |  |
| Internal web by IP | http://192.168.10.10 | HTTP client-server works internally. |  |
| Internal web by name | http://intranet.archstudio.local | DNS resolves the internal name to its IP. |  |
| External server ping | ping 203.0.113.10 | Router connects the office LAN to the external network. |  |
| External web by IP | http://203.0.113.10 | Client can access an external service. |  |
| External web by name | http://cloud.archstudio.example | DNS resolves the external service name. |  |

| | |
|---|---|
| TCP/IP | All device communication |
| IP addressing | Every device has a unique IP |
| Subnet mask | 255.255.255.0 on both networks |
| Gateway | Client PCs use 192.168.10.1 |
| DNS | intranet.archstudio.local etc. |

**Protocols to explain**

**TCP IP:** The TCP/IP model is the practical frame ware that powers the internet. Its combination of five layers psychical layer, data link layer, internet layer, transport layer and application layer.

**IP Addressing:** Ip address is a unique string of numbers assigned to any device connected to the internet or local area network.

**Subnet musk:** Subnet is a 32-bit number that splits an IPv4 address into a networking ID and host ID.

**Gateway:** is the router local IP address that connects the client PC to the internet.

**DNS:** is the domain name system that translates the domain name, for example (intranet.archstudio.local etc.) into Ip address 192.168.10.121.

LAN vs External Network

A LAN (Local Area Network) is a private network covering a single area, for example a home, office, etc. - it connects devices like computers, phones and printers. However, an external network, also referred to as a WAN (Wide Area Network), connects multiple LANs together to allow devices to access data/servers/websites over very large distances, even worldwide.
A client-server is where the client (your device) asks for data/services from a server over a network. The process is essentially requesting something, server processing this request and choosing whether to give it, then responding by giving the service or not.